

General Advice for Physics Olympiad Training

ktanikun

2026 年 5 月 29 日

Translation note

This document was translated from Japanese into English using an LLM. The translation has been reviewed lightly for terminology, readability, and preservation of the original intent.

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Preface

This document organizes advice on how to approach the Physics Olympiad, mainly from the viewpoint of maximizing one's score in the exam.

Of course, the value of the Physics Olympiad is not limited to points or medals. Studying physics deeply, thinking about unfamiliar phenomena with one's own mind, interacting with high school students from around the world, and continuing to learn science after the competition all have great value beyond exam results. Still, as long as the Physics Olympiad is a competition, one cannot avoid thinking about how to score points in the theoretical and experimental exams.

Therefore, this document will not go deeply into value judgments such as whether one should enjoy physics or aim for a medal. Instead, it will describe concretely what abilities are needed to score highly in the Physics Olympiad, how to train those abilities, and how to use them during the actual contest.

This does not reduce the Physics Olympiad to a mere point-scoring game. Rather, if we break down the process of scoring points in the exam, we find many abilities that are also important for learning physics: physical knowledge, modeling ability, calculation skill, experimental skill, answer-writing skill, and the ability to handle the real exam. The viewpoint of maximizing the score is one way to analyze these abilities concretely instead of leaving them vague.

Part I. Advice for Students

1 Skills Required in the Physics Olympiad (Theory Exam)

The theoretical exam in the Physics Olympiad is not simply a test of difficult knowledge. Within a limited time, you must read unfamiliar problems, build physical models, carry out calculations, and present your reasoning in a form that graders can follow. Thus, what matters is not only whether you know something, but whether you can keep solving on the spot.

How to think about the score

The score in the theoretical exam can be thought of as

$$\text{score} = \text{solved points} - \text{lost points}.$$

In other words, to increase your score, you first need to increase the total value of the subproblems you can solve. If your hand stops during a problem, you lack the ability to keep moving forward. If you run out of time, you lack processing speed. In most cases, your exam score is bounded above by this “solved points” side. Lost points are mistakes in problems that you essentially solved.

However, one should be a little careful about how to think about lost points. If you write a consistently correct approach, you are often saved in later subproblems by carry-forward grading. Conversely, if you write an answer that is physically false, or whose dimension or order of magnitude is clearly wrong, it becomes harder for the later reasoning to receive credit. Roughly speaking, “nonessential mistakes” are penalized only mildly as long as they do not damage the essential physics.

Below is one way to classify the abilities needed to score points in the theoretical exam.

1.1 Physics knowledge

Physics knowledge is the foundation for understanding the background of a problem and selecting the laws that should be used. It is of course necessary to know the basic laws of mechanics, thermodynamics, electromagnetism, waves, and modern physics, in other words the content of the IPhO syllabus. If this foundation is not yet solid, you need to start there. More concrete details will be discussed in the next section.

That said, merely memorizing formulas is not enough, and it is not necessary to memorize every formula perfectly. Very complicated formulas are usually not that important. What matters is that you understand the ideas well enough to use them yourself. Knowledge is not useful here if it remains just knowledge.

At a higher level, especially for students aiming for an IPhO gold medal, knowledge really does matter. At least at present, many Physics Olympiad problems are built around patterns that can be called standard. For a sufficiently trained student, most problems are not truly unseen; they ask for familiar procedures in a slightly different setting. If a problem feels familiar, you can proceed smoothly without missing the essential point, even if the setup is somewhat different.

There are also many “new-looking” problems based on some physical phenomenon. In such cases, simply

knowing related background knowledge can be a large advantage. In that sense, it is meaningful to know physics beyond the IPhO syllabus, as well as more practical physics knowledge.

1.2 Understanding the problem

Understanding the problem means reading the physical situation from the statement and clarifying what exactly should be solved. If this ability is weak, you may get stuck before writing any equations even if you have enough knowledge. In Physics Olympiad problems, the sequence of subproblems itself often acts as guidance. In practice, two abilities are especially important: reading the intention of the problem setter's guidance, and avoiding misreadings of the statement.

1.3 Choosing an approach

This is the ability to decide what to do as the first move. In recent IPhO exams, the point values of individual subparts are often divided into small pieces. In such cases, once you find one thing to do, the rest may simply be calculation. Many problems are essentially "do what you are told" problems. The ability not to go blank at that moment is what I mean here. It is a kind of matching ability: from the diagram, decide to calculate, differentiate, integrate, write an equation of motion, formulate a conservation law, and so on.

On the other hand, in IPhO or APhO subproblems with larger point values, say 1.5 points or more, you are strongly required to decide the approach by yourself. In competitions such as EuPhO, creativity may be required even more. Sometimes it is not enough to possess many existing solution patterns; you may need to derive a clever solution yourself.

1.4 The ability to keep solving

The ability to keep solving is the ability to carry calculations and logic through once you have chosen an approach. In other words, it is the ability not to let the solution collapse halfway. For example, after setting up an integral, you need to actually compute it without getting stuck.

It is also quite important in recent IPhO exams to connect one equation to the next using the guidance of the subproblems, or to move on using a given result even if you could not solve the previous subproblem. This kind of resilience allows you to collect many partial points even when you cannot solve a problem completely.

1.5 Processing speed and accuracy

If your goal is to score more than 20 points in theory, time is extremely tight. There are three problems in five hours, and the number of subparts can exceed 30, so there is simply a lot to do. Time is always insufficient.

Therefore, even if you have the ability to solve the problems, your score will not grow if your processing is slow. You need to carry out basic calculations quickly, avoid hesitation in standard manipulations, and finish easy parts in a short time. Deciding not to spend too much time on a hard subproblem is also part of processing speed. In addition to calculating faster, there are also techniques for organizing the work so that

even heavy calculations become cleaner.

On the other hand, accuracy is needed to convert a correct approach into points. Sign errors, coefficient errors, inconsistent units, missed cases, and wrong orders in approximations all reduce your score. Once you obtain an answer, you should have the habit of checking dimensions, signs, and whether the answer behaves naturally in special limits. However, it is also bad to overemphasize accuracy so much that your hand stops. What matters is to avoid fatal breakdowns while maintaining an acceptable speed within the exam time.

1.6 Answer-writing skill

An answer sheet is not your personal calculation memo; it is a document for communicating your reasoning to the graders. You need to state the symbols you use, the coordinate system, the positive direction, the laws applied, and the approximation conditions. Do not write only equations; add short explanations where necessary. Show final results clearly, and even when you cannot solve a problem completely, leave correct intermediate equations.

Especially in international competitions, it is important to write an answer whose flow can be followed by graders from the host country. If graders can see what the student is trying to do, the answer is more likely to receive partial credit even if there is a small calculation error along the way.

1.7 Performance in the actual exam

This is the ability to maximize your score even when things do not go according to plan during the exam. The difficulty of the problems varies greatly from year to year, and in international competitions there can also be external factors. Still, you must not let that break you. That is the kind of ability meant here.

A student who is strong in the exam is not necessarily a student who solves everything perfectly. Rather, it is a student who can keep accumulating points even in unexpected situations. Since you only get a few real chances in life, it is more valuable to be able to score decently even on a bad day than simply to have a high ceiling on a good day.

Summary of this chapter

The abilities needed in the theoretical exam cannot be summarized by saying simply that someone is “good at physics.” At minimum, they should be separated into physics knowledge, problem-understanding ability, approach-selection ability, the ability to keep solving, processing speed, accuracy, answer-writing skill, and real-exam performance. When your exam score does not improve, it is better to break down which ability is lacking than to conclude vaguely that you cannot do physics.

2 Theory Preparation: Long-Term Plan

To be written

This chapter will discuss how to build a long-term plan for theoretical preparation: strengthening the basics, practicing by field, working through past problems, and moving toward full-set practice.

From here on, I mainly describe things that can be done before the competition.

2.1 What matters in past-problem practice

For theoretical preparation in the Physics Olympiad, practicing past problems is one of the most important forms of training. However, it is not appropriate to set the goal of past-problem practice merely as hoping that the same problems will appear in the real exam. In practice, past-problem practice has several purposes.

- Becoming familiar with the style of IPhO and APhO problems.
- Practicing time allocation and decisions about when to retreat in the real exam.
- Understanding, through the Marking Scheme, how partial credit is awarded.
- Measuring your current ability.
- Improving physical knowledge and calculation skill through problem solving.

Among these, the especially important goals right before the competition are familiarity with the format, establishing exam strategy, and learning how to earn partial credit. Therefore, past-problem practice should not just mean solving problems; it is important to do it in a format close to the actual exam.

2.1.1 Distinguish full-set practice from single-problem practice

Past-problem practice can be broadly divided into two types.

- Full-set practice: solving three problems in five hours.
- Single-problem practice: solving one problem as an exercise.

These two have different purposes. Single-problem practice is done to understand one problem deeply and acquire physical knowledge and solution methods. Full-set practice, on the other hand, is done to practice how to spend time, where to retreat, and where to collect partial points under the same time limit as the real exam.

After team selection or shortly before an international competition, single-problem practice alone is not enough. The theory exam requires you to process several large problems in five hours, so you should do several full-set practices with the real exam time in mind, although it can be difficult to secure such long blocks of time.

2.1.2 Treat full-set practice like the real exam

When doing full-set practice, it is best to create conditions as close to the real exam as possible. Concretely, prepare the following.

- Five uninterrupted hours and an environment where you can concentrate.
- The problem statements, Answer Sheets, and blank Working Sheets.
- Writing tools, a calculator, and a timer.
- The official solution and the Marking Scheme.

If possible, solve the problems on paper. Reading a problem on a screen and writing answers while turning printed pages feel quite different as information-processing tasks. Also prepare enough Working Sheets. In the real exam, it is important to leave intermediate equations and reasoning on paper that can be graded, so you should get used to that format during practice. Answer Sheets are not strictly necessary for practice, but in the real exam you will write final results there.

During full-set practice, do not merely check whether you can solve the problems; try to maximize your score as if it were the real exam. Even if you cannot solve one subproblem, you do not need to abandon the whole problem. You may be able to earn partial credit by assuming a previous result and moving on, introducing a symbol for an unknown quantity, writing a conservation law that seems relevant, or doing dimensional analysis. It is better to struggle as much as you can and challenge yourself to see how many points you can get. What you cannot do in practice, you cannot do in the real exam.

2.1.3 Read the Marking Scheme to understand the grading structure

After past-problem practice, grade your work using the official solution. At that time, you should check the Marking Scheme. If you only read the solution, you can learn the correct solution, but it is hard to know which intermediate equations actually receive points and how much various mistakes are penalized. Marking Schemes can be found on sites such as [phoXiv](https://arxiv.org/abs/1905.09275).

Reading the Marking Scheme reveals things such as the following.

- Points may be given for the approach or for setting up equations.
- Even if the final result is wrong, intermediate equations may receive partial credit.
- If an incorrect answer from an earlier part is used consistently, later subproblems may still receive credit; dimension errors and physically unnatural equations tend to cause large losses.
- Figures, graphs, axis labels, units, and significant figures may also have points or penalties assigned to them.

Therefore, in reviewing past-problem practice, do not stop at saying that you solved or did not solve a problem. Check how many points your answer would receive under the Marking Scheme. Then think about how you could have earned more points with the same level of understanding.

2.1.4 Understand where you stand

Since you have gone to the trouble of grading your work, you can look at resources such as [IPhO Timeline](#) to see what scores were achieved in that year's competition. This helps you understand your current position. For example, if you scored 10 points, you can see whether that was close to the gold-medal line or the silver-medal line in that year. It may also be interesting to ask your teammates about their scores. You can evaluate how close you are to your goal.

2.1.5 Advanced: analyze how many points you lost

When reviewing past-problem practice, in addition to understanding the problem to the end, you need to analyze how you lost points. For example, for a problem on which you scored only 5 points, do not simply think, "It was difficult." Think about how you could have gotten closer to 7, 8, or 10 points. The causes of lost points can be classified as follows.

- You lacked physics knowledge.
- You did not understand the situation in the problem statement correctly.
- You could not decide on an approach.
- Your approach was correct, but you could not carry out the calculation.
- You ran out of time.
- You made calculation, sign, or dimension errors.
- You did not leave enough explanation or intermediate work in the answer.

Once you classify the causes, it becomes clear what you should train next. If the issue is lack of knowledge, return to field-by-field review. If the issue is time, increase full-set practice. If you make many calculation errors, you need to make dimension checks and special-case checks into habits.

2.1.6 In single-problem practice, understand the problem to the end

In single-problem practice, emphasize deep understanding of the problem itself rather than score maximization as in the real exam. After spending a fixed amount of time on one problem, follow the structure through to the final part, even if you need to read the solution.

In particular, keep the following points in mind.

- What is the central physical model of the problem?
- Which conservation laws, symmetries, and approximations were essential?
- Why were those variables or that coordinate system chosen?
- Which subproblems were preparation for later arguments?
- How much partial credit could you have earned in the real exam?

Unlike full-set practice, single-problem practice is training for depth of understanding rather than strategy under time pressure. Therefore, do not end the review simply by reading the solution. Ideally, you should

reach the point where you can reconstruct the solution by yourself later.

2.1.7 Practice past problems with a plan

Trying to solve a large number of past problems before the competition is not very effective if you cannot digest them. What matters is not the number of years or problems solved, but what you learned from each practice.

Especially in the final period before the competition, it is helpful to separate goals as follows.

- Practice time allocation through full-set practice.
- Learn the grading sense by using the Marking Scheme.
- Strengthen weak fields and standard techniques through single-problem practice.
- Re-solve problems so that once-solved problems become reliably scoreable.
- Train small techniques such as dimensional analysis, numerical calculation, calculator use, and answer writing.

Past-problem practice is not for merely consuming problems. It should be used to find your weaknesses, improve your exam strategy, and practice writing answers that are actually graded, all in order to score higher in the real exam.

2.1.8 Connect the practice to the next step

If you are aiming for an IPhO medal, you do not need to review every part of every problem. Although it depends on the year, a good target is probably around 10 to 15 points. First, check whether you solved the early part of Part A, where the basics are tested. If not, review the relevant basic knowledge. If you could do that, then, while looking at the solution, try again up to the parts in each large problem that carry relatively large point values. During this process, you may find gaps in your knowledge. If you can solve the large, central parts of each problem, you are probably close to 15 points.

If you are aiming for an IPhO gold medal, you should review almost everything in IPhO past problems and aim for roughly 20 to 25 points. The review method is mostly what I wrote above. However, if you still do not understand a problem after reading the explanation, the problem itself may have issues, so you may want to check resources such as [IPhO Errata](#), or ask teammates or an LLM. Also, it is important to touch many problems beyond IPhO past problems; short problems, for example Kevin-style problems, are also worth using.

3 Toward the Theory Exam

Advice for the theory exam

Here I divide advice for the actual theoretical exam into three categories: strategy during the exam, techniques on the answer sheet, and external management. These are also worth keeping in mind during past-problem practice.

3.1 How to behave during the exam

Look over the whole exam first

Immediately after the exam starts, it is better not to begin solving Problem 1 right away. Your engine is not yet running. If you start immediately, things often do not go well and you may become anxious. For the first few minutes, look over the whole exam and check each problem's field, point value, and so on. This lets you think from above about which problem to start with, which problems seem heavy, and where partial credit may be available, while also calming the initial disturbance of the exam.

Read all parts before entering a problem

Before working on one large problem, read not only the introduction but all of its subparts. In Physics Olympiad problems, later subparts sometimes tell you the meaning of earlier ones. Also, even if you cannot solve an earlier part, a later part may explicitly state the result you need. Reading all parts also helps you understand where you currently are. When you get stuck, you can judge whether the thing you are solving now is an important part or merely the beginning.

Sometimes one large problem contains two completely independent questions. If you fail to notice this, you may lose a large number of points that were actually available. As an advanced point, if you understand the dependency structure and point allocation, you can judge that a part is only a branch and can be skipped even if the problem is not completely independent.

Move on even if you cannot solve the previous part

When you cannot solve an earlier subproblem, it is easy to give up on the entire large problem. However, later subproblems can often be solved by assuming the result of the previous one. If the previous result is given, use it. Even if it is not given, it may be worth introducing a suitable symbol and moving on.

Before leaving a problem, leave traces and struggle

Before leaving a large problem, it is good to leave some trace for every subpart. Even if you cannot solve it completely, write down the law you would use, a diagram, variable definitions, dimensional analysis, or limiting behavior. This becomes a restart point when you come back later, and it may also become partial credit.

In the theoretical exam, what matters is not how many problems you solved completely, but how many points you collected from each subproblem. Sometimes it is better to reliably collect available parts from several problems than to solve half of one difficult problem.

Do not panic if you fall behind; skip after stopping; do not leave a whole problem untouched

In the Physics Olympiad, people sometimes say “one point per ten minutes.” Since the theory exam is out of 30 points, this is the average pace. But you cannot maintain this pace in every situation. You will fall behind at some point, because if you could keep that pace perfectly, you would score full marks. Therefore, there is no need to panic just because you are behind. The rough pace of 100 minutes per problem is also usually exceeded.

So how should you manage time? It is enough to follow two rules.

First, do not let your hand be completely stopped for more than about five minutes. It is dangerous to keep thinking for a long time while writing nothing. In particular, if time is passing while you do not know what to write, you should leave that subproblem once. At that moment, your score is not increasing. Especially in an IPhO-style exam, there are many other problems you should work on rather than thinking too long about one point. It is good to decide your own rule, for example to skip a subproblem after about five minutes of being stuck.

Sometimes this happens because your concentration has dropped. In that case, it is good to refresh yourself in some way.

Second, manage your time so that you touch every large problem at least a little before the exam ends. The early part of another problem is often easier and worth more than the late part of a difficult problem. If you barely touch one whole large problem, you may lose many important points. For example, even in a worst-case plan, you might leave Problem 1 with three hours remaining and leave Problem 2 with one hour remaining.

3.2 Techniques useful in the exam

Draw a diagram

Rather than thinking only from the text of the problem, it is better to draw a diagram first. If you draw the directions of forces, coordinate axes, angles, distances, current directions, light paths, heat flow, and so on, the laws to use become easier to see. A diagram is useful not only for your own understanding but also as part of the answer.

Before writing a complicated differential equation, suspect a conservation law

Before setting up a complicated differential equation, consider whether a conservation law can be used. If energy, momentum, or angular momentum is conserved, you may avoid solving a differential equation. However, if there are nonconservative forces or external forces, you need to consider which conservation laws are broken.

Look for symmetry

Symmetry is powerful information. If there is left-right symmetry, rotational symmetry, translational symmetry, time-reversal symmetry, and so on, the directions of forces, the form of fields, conserved quantities, and boundary conditions are strongly constrained. Noticing symmetry can greatly reduce the amount of calculation.

Use dimensional analysis and keep familiar blocks together

Always check whether the dimension of your answer is correct. An equation with the wrong dimension is almost certainly wrong. Also, even when you do not know where to start, dimensional analysis may let you predict the form of the answer.

Moreover, if you expand expressions unnecessarily, their structure becomes hard to see and calculation errors increase. Quantities with physical meaning should be kept together as much as possible. For example, energy, time constants, and angular frequencies should be managed in familiar grouped forms. This makes the meaning of intermediate equations easier to understand and reduces mistakes.

Introduce symbols

If a frequently appearing block has some meaning and is not defined in the problem, it is good to name it with a symbol such as E_0 , τ_0 , or ω_0 . This helps both organization and speed. However, because the grader will not know your notation otherwise, you must always define it.

Also, even if an unknown quantity appears in the middle of a solution, you do not need to stop there. You can denote it by a symbol such as A or x and continue with later equations. It may cancel out later, or it may be given in a later subproblem.

Write the approach in the answer; use dimensional analysis when you have nothing else

If you write only equations, then once there is a small mistake it becomes unclear what you are doing. Short explanations such as “Using conservation of energy” or “Since no external force acts on this system, momentum is conserved” make it easier to receive partial credit.

Also, even when you cannot think of anything, trying dimensional analysis to predict the form of the answer is a standard but very effective move. Even if you cannot solve that subproblem correctly, opening a path to later subproblems is very valuable.

Check dimensions, signs, and coefficients separately; look at limits

When checking your work, check dimensions, signs, and numerical coefficients separately. An expression can have the correct dimension but the wrong coefficient, or the right coefficient but the wrong sign. For signs in particular, return to the positive direction of the coordinate system and the directions of the forces. Checking by physical meaning is also important.

Once you have an answer, check whether it behaves naturally in limiting cases. Look at limits such as

mass approaching 0, resistance going to infinity, small angles, or very large distances. If the answer is clearly strange in a limit, there is likely a mistake somewhere.

3.3 External management

Use Working Sheets generously

Even work that began as scratch work may lead to partial credit if it remains on paper that can be graded. A *Working Sheet* is not something to save; it is a place to show your thinking to the graders. It is better to separate it by problem and use plenty of space. In any case, writing a lot is advantageous, and it can also widen the possibility of partial credit when your own country's leaders look at your work.

If you think you may run out of Working Sheets, ask for more a little early. It may take time before they actually arrive.

Keep your desk organized

In a five-hour exam, the desk can become cluttered with problem statements, answer sheets, *Working Sheets*, writing tools, a clock, water, snacks, and so on. Time spent searching for papers is wasted, and it interrupts thought. Keep only the papers for the large problem you are currently solving near you, and move unused papers to the side.

Practice with the writing tools in mind

In a competition, you may not be able to use your favorite writing tools. You may need to write for a long time with a ballpoint pen or a specified pen. It is good to practice writing answers in an environment somewhat close to the real one.

Prepare a watch

The clock in the exam room may not be easy to see. You should be prepared to check the time yourself. However, instead of becoming anxious every time you look at the clock, check your progress at predetermined moments.

Manage water, sugar, and clothing

In a five-hour exam, maintaining concentration also affects the score. First, you should drink water frequently. Dehydration greatly reduces concentration. If the air conditioning is weak, you may be close to dehydration before noticing it, so be especially careful. In most cases, water is provided without limit.

Also bring snacks that you like to the exam room. Sugar helps maintain concentration, and the snacks provided at the contest often do not suit your taste. It also helps keep your mood stable.

Finally, wear clothing that can handle a room that is cold or hot. Overall, physical condition management is not a matter of spirit; it is a condition for maintaining processing speed and judgment.

Do not fear restroom breaks or short rests

If your concentration has collapsed, staying seated will not easily increase your score. If necessary, it may be better to take a short restroom break or reset your posture. Standing up and stretching inside your booth can also be fine. However, if you rest, keep it short and decide what to restart from when you return.

Decide your exam-day routine beforehand

If you first start thinking about time allocation or paper management during the actual exam, it creates unnecessary cognitive load. Decide in advance how you will look over the whole exam, how you will use the *Working Sheet*, when you will check the time, and what you will check at the end.

4 Experimental Preparation

To be written

This chapter will cover measurement, uncertainty, graphs, fitting, experimental report writing, and planning during the real exam.

Here I briefly summarize what kind of preparation can be done at home or in a training camp for the experimental exam.

4.1 Preparation

Use past experimental problems

Using past experimental problems, you can imagine what kind of measurements are required and try analyzing the data provided in official solutions.

Learn how to use basic equipment

Learn how to use equipment that often appears in IPhO experiments. Examples include ammeters and voltmeters, oscilloscopes, and optical setups. The best way to learn these is to actually touch them.

Understand the theoretical background

Even if an experiment simply tells you to do something, understanding why it works gives you more flexibility in the real exam. For example, if you understand the theory behind circuits, optics, and oscillations, you can understand the meaning of the measurements more deeply.

Learn data-analysis methods such as uncertainty analysis and linearization

As experimental knowledge, data handling, uncertainty analysis, and linear fitting are important in the IPhO. You need to study these as seriously as theory and become able to use them in practice.

Practice writing tables and graphs

There are rules for writing tables and graphs. Following them gives graders a better impression and reduces unnecessary penalties. In doing this, it is good to check carefully what the Marking Scheme looks at.

Practice using the calculator efficiently and accurately

It is also important in experiments to use a scientific calculator well. You may not actually be using its convenient functions, so it is worth investigating them. Accuracy is also important; it is sad if correctly analyzed data becomes wrong during calculation.

Get access to experimental apparatus

In the end, the most effective preparation is to have teachers let you use real equipment during a training camp and to attempt past problems with it. If you do this in a group, it is important to stay calm and work through the deeper parts of the problem.

If this is not possible, ktanikun's site has something called experiment drill. It lets you practice planning, theory, and analysis other than actually measuring with apparatus, so I hope you try it.

4.2 Advice for the real exam

Do not start measuring immediately

This is important: you need to understand deeply what you are being asked to do. If you anxiously start touching the apparatus, the operation is usually wrong, and in the worst case you may break the equipment.

If the measurement does not work

Stay calm and investigate the cause. The apparatus may be assembled incorrectly, or a component may be broken. In such cases, do not hesitate to call the exam supervisor or Technical Support. They may be able to help.

5 My Own Path: What I Worked On as a Student

To be written

This chapter will discuss the materials, practice, reflections, periods of growth, and periods of stagnation from my own time as a student.

Part II. Advice for Tutors and Mentors (for other countries' coaches)

6 Introduction to Japan's Training and Selection System

To be written

This chapter will introduce the flow of selection, training, and representative selection in the Japanese Physics Olympiad system.

7 Coaching Techniques

To be written

This chapter will cover practical techniques for mentors, such as problem explanations, grading feedback, interviews, and designing full-set practice.

Although this was not full-fledged coaching, I will introduce what I did with my teammates shortly before the contests as the self-proclaimed team captain for EuPhO 2024 and IPhO 2025.

7.1 Toward EuPhO

Japan rarely participates in EuPhO. It happened that the IPhO delegation was canceled that year, and we went to EuPhO as an alternative. Since it was decided suddenly, there was not much know-how, so we did the following.

- We solved problems from *Puzzling Physics Problems* and *More Puzzling Physics Problems*, and shared problems that were especially interesting or useful, because outside EuPhO past problems, those books seemed close in style.
- We held an online team meeting once a week and shared insights from the problem practice above and information we had collected about EuPhO, such as problem format, grading method, and other systems. Explaining a favorite problem was also fine.
- In each meeting, we also assigned one year's EuPhO theory past problems to be solved by the next week. In the following meeting, we shared impressions of that past exam, and difficult problems were explained by people who had solved them.
- We shared written solutions on Google Drive and practiced moderation. Concretely, we graded other people's answers while looking at the Marking Scheme, and the person whose answer was graded argued in English to raise their score. We did this because in EuPhO, moderation is conducted by students themselves in English, so practice was necessary.

7.2 Toward IPhO

In my final year, 2025, I was selected for IPhO and the participation was confirmed. We seriously worked toward the goal of all members winning gold medals. We looked not only at theory but also at experimental preparation that could be done at home. The following mainly refers to the period after APhO and before IPhO.

- First, we reflected on APhO.
- Next, we held a team meeting once a week and made concrete goals and a plan for past-problem practice. We decided to assign one common year of past problems per week, because some members were also representatives for IChO and others were busy with school exams.
- In the meetings, we mainly reflected on the past problems. For problems we could not solve or did not understand well, we discussed in detail what was strange and resolved the issue. Since sometimes the problem itself is wrong, it was good to be able to share that.
- Since we knew the model of the calculator that would be distributed in the exam, several of us bought it, practiced using it, and discovered and shared useful functions.
- For experimental preparation, I created various exam strategies myself. In particular for analysis, I practiced a lot using experiment drills. I also taught only the techniques to teammates who had little time, and shortly before the contest we gathered at my house to practice writing tables and graphs.
- This was unrelated to the exam, but we also researched various things about the travel destination, Paris.

In particular, the weekly team meeting helped maintain the team's motivation. Also, since Japan placed second in the experimental exam by average score after using the experimental preparation I had designed, I think it had an effect.

8 Closing Remarks

To be written

Write some kind of summary here.